

QIIME 2 16S rRNA Pipeline

Technical Review

Amplicon microbiome pipeline · `scripts/qiime-console.py`

The problem

- Raw multiplexed Illumina 16S reads → **who is in each sample, and how do communities differ?**
- Needs to be **reproducible** (provenance, checksums) and **shareable** with collaborators who may re-analyze.
- Historically: scattered shell scripts + a legacy QIIME 1 CWL path, no single driver, no packaged deliverable.

The goal

- One driver (`qiime-console.py`) from raw reads → taxonomy + diversity.
- EMP paired-end protocol, **DADA2 ASVs** (exact sequences, not 97% OTUs).
- A curated, checksummed **deliverable** bundle for hand-off.
- Documentation for PIs, users, developers, maintainers.

Pipeline at a glance

```
setup → import → demux → dada2 denoise  
      → phylogeny (+ taxonomy/classify/barplot)  
      → diversity (core-metrics, alpha/beta significance, rarefaction)  
      → relative abundance (collapse L1-7)  
      → package (deliverable/ + .tar.gz + MANIFEST)
```

Full Mermaid diagram + tool/output tables: [PIPELINE.md](#)

⚠ Taxonomy runs *inside* `run_phylogeny_analysis`, **before** diversity.

Inputs

- **Raw reads:** `Undetermined*_{R1,R2,I1}*.fastq.gz` (forward / reverse / barcode), staged and renamed by `setup`.
- **Mapping file:** tab-delimited QIIME metadata; `#Sample ID` + `BarcodeSequence` (12 bp Golay).
- **Classifier:** Greengenes 13-8 515F/806R naive-Bayes, fetched by `setup` (pinned to `sklearn-1.4.2`).

Per-stage walkthrough

Stage	Tool	Produces
Import	<code>tools import</code>	EMP paired-end artifact
Demux	<code>demux emp-paired</code> + summarize	per-sample reads + QC
Denoise	<code>dada2 denoise-paired</code>	ASV table, rep-seqs, stats
Phylogeny	<code>align-to-tree-mafft-fasttree</code>	rooted tree
Taxonomy	<code>classify-sklearn</code> + <code>taxa barplot</code>	taxonomy + bar plots
Diversity	<code>core-metrics-phylogenetic</code> + significance	alpha/beta + PCoA
Rel. abund.	<code>taxa collapse</code> L1–7	rank tables

Outputs: exported vs internal

Exported (in the deliverable): ASV table, rep-seqs, denoising stats, rooted tree, taxonomy, taxa bar plots, demux QC, `core-metrics-results/`, `alpha-rarefaction-results/`, `metadata.tsv`, `MANIFEST.txt`.

Internal (working products, not shipped): `paired-end-demux.qza`, demux artifacts, alignments, unrooted tree, `phyla-table.*` / `rel-phyla-table.*`.

Split matches `DELIVERABLE_FILES` / `DELIVERABLE_DIRS` in code.

Packaging & provenance

- `package` builds `output/deliverable/` **and** `<run>-deliverable.tar.gz`.
- Includes `.qza` **data** artifacts (for downstream re-analysis) + `.qzv` viewers + metadata.
- `MANIFEST.txt` : md5 checksum + size per file, run params, QIIME version.
- **Per-artifact provenance**: every `.qza` / `.qzv` embeds its full action graph + citations → <https://view.qiime2.org> → Provenance tab.

QC & interpretation

- `demux-full.qzv` → choose truncation lengths & sampling depth.
- `stats-dada2.qzv` → reads surviving filter/merge/chimera.
- ⚠ `--p-sampling-depth` **default 20000** drops any shallower sample — set it from the summaries, don't accept the default.
- Diversity + PERMANOVA answer *richness*, *evenness*, and *between-group* differences.

Reproducibility

- Deterministic driver; parameters recorded in `MANIFEST.txt` .
- QIIME provenance travels **with** every artifact.
- Checksums verify bundle integrity on transfer.
- Docs stamped to a commit; `check_docs_tables.py` guards table/code drift.

Known issues & roadmap

- **Fixed recently:** `run demux` works; `--barcode-column-name` honored; metadata headers trimmed; shell classifier name corrected.
- **Tracked:** remaining metadata hardening (issue #5).
- **Roadmap:** QIIME 2 CWL port (`wf.1.0.cwl`); retire legacy QIIME 1 CWL; Python driver as single source of truth.

Summary

- End-to-end QIIME 2 ASV pipeline in one driver.
- Curated, checksummed, provenance-rich deliverable for collaborators.
- Documented for every audience; tables kept honest against the code.

Docs: `docs/` · **Reference:** [PIPELINE.md](#)